No natural selection- All alleles are fit to survive and reproduce.

If any one of these assumptions were not met, the population will not be in Hardy-Weinberg equilibrium. Only if the allele frequencies changes from one generation to the other, evolution will take place.

6.8 Origin and Evolution of Man

Mammals evolved in the early Jurassic period, about 210 million years ago (mya). Hominid evolution occurred in Asia and Africa. Hominids proved that human beings are superior to other animals and efficient in making tools and culture.

The earliest fossils of the prehistoric man like *Ramapithecus* and *Sivapithecus* lived some 14 mya and were derived from ape like *Dryopithecus*. *Dryopithecus* and *Ramapithecus* were hairy and walked like gorillas and chimpanzees. *Ramapithecus* is regarded as a possible ancester of *Australopithecus and* therefore of modern humans. They were vegetarians **(Fig 6.10)**.

Australopithecus lived in East African grasslands about 5 mya and was called the Australian ape man. He was about 1.5 meters tall with bipedal locomotion, omnivorous, semi erect, and lived in caves. Low forehead, brow ridges over the eyes, protruding face, lack of chin, low brain capacity of about 350 – 450 cc, human like dentition, lumbar curve in the vertebral column were his distinguishing features.

Homo habilis lived about 2 mya. Their brain capacity was between 650 – 800cc, and was probably vegetarian. They had bipedal locomotion and used tools made of chipped stones.

Homo erectus the first human like being was around 1.7 mya and was much closer to human in looks, skull was flatter and thicker than the modern man and had a large brain capacity of around 900 cc. *Homo erectus* probably ate meat *Homo ergaster* and *Homo erectus* were the first to leave Africa.

Neanderthal human was found in Neander Valley, Germany with a brain size of 1400 cc and lived between 34,000 - 1,00,000 years ago. They differ from the modern human in having semierect posture, flat cranium, sloping forehead, thin large orbits, heavy brow ridges, protruding jaws and no chin. They used animal hides to protect their bodies, knew the use of

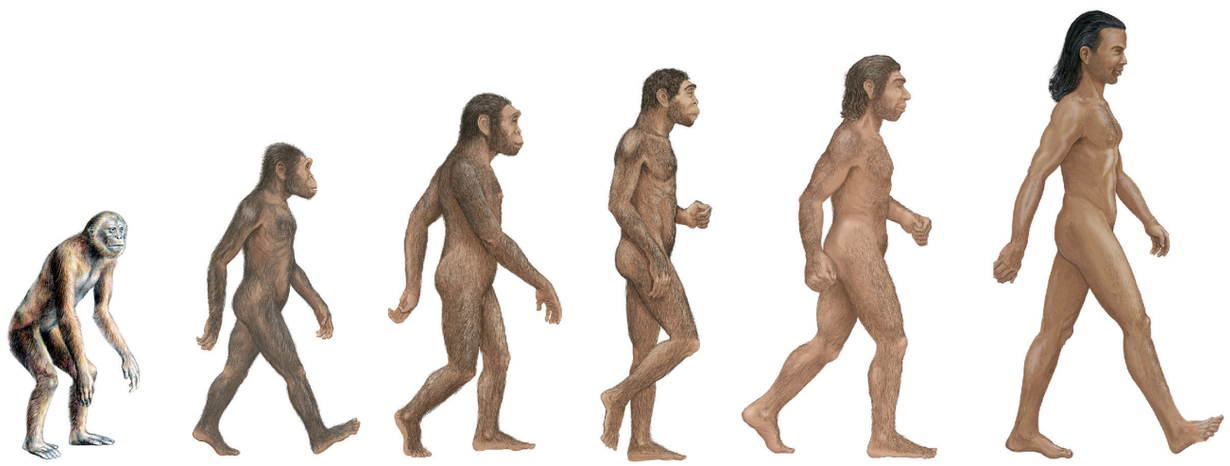


Fig 6.10 Evolution of Man

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